

Theta Closure in Modal Triplet Theory V: Weak-Angle Round Trip and the Non-Circularity Criterion

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Abstract

We reassess whether the weak mixing angle provides a redundant test of the selected MTT gauge profile. Let $r_{21} = I_2/I_1 = g_1^2/g_2^2$, with $g_1 = \sqrt{5/3} g_Y$. Then $\sin^2 \theta_W = 3r_{21}/(5 + 3r_{21})$ at the same scale and in the same scheme. For the selected SMDR profile at $Q = M_t$, this gives $\sin^2 \theta_W = 0.2346644 \pm 0.0000433$. Because r_{21} was itself obtained from the measured (g_Y, g_2) profile, this equality is an exact algebraic round trip, not a held-out prediction. Supplying an absolute g_2 from (G_F, m_W) does not remove that circularity. A genuine redundancy test requires an MTT source theorem selecting r_{21} without electroweak gauge data, followed by independent common-scheme transport.

Revision note for this edition

Supersedes. *Theta Closure in Modal Triplet Theory V: Redundant Determination from Gauge Couplings and the Weak Mixing Angle*, first edition.

Reason. The weak angle was presented as an independent precision test even though it was algebraically reconstructed from the same measured gauge profile, and the scale used the retired few-TeV crossing.

Resolution. Version 2 derives the same-scheme round-trip formula, propagates its uncertainty at $Q = M_t$, and states an explicit non-circularity criterion for any future test.

Retained result. The weak angle remains an exact consistency identity for a supplied gauge-ratio profile.

Remaining boundary. A held-out test requires MTT to select r_{21} without electroweak gauge data and then transport it independently.

1 Central picture and roadmap

The central picture is that the weak mixing angle describes how the neutral electroweak gauge fields are rotated into the photon and Z directions. In a fixed renormalization scheme and at one scale, that angle is determined by the same pair of couplings (g_Y, g_2) . It is therefore not a new datum when the ratio of those couplings has already been used.

The argument has three steps:

$$(g_Y, g_2) \longrightarrow r_{21} = \frac{5g_Y^2}{3g_2^2} \longrightarrow s_W^2 = \frac{3r_{21}}{5 + 3r_{21}}.$$

Section 3 proves that the composition returns $g_Y^2/(g_Y^2 + g_2^2)$ identically. Section 4 explains why adding an absolute $SU(2)$ normalization does not create a second ratio. Section 5 states the data-separation conditions required for a genuine held-out test.

2 Selected inputs and scope

Paper I adopts the full-Standard-Model $\overline{\text{MS}}$ profile transported by SMDR v1.3 to

$$Q = M_t = 172.5590883453979 \text{ GeV}.$$

Its relevant entries are

$$g_Y = 0.3585945042 \pm 0.0000307251, \quad g_2 = 0.6475986708 \pm 0.0000287665,$$

or $g_1 = \sqrt{5/3} g_Y$ in GUT normalization. The associated overlap ratio is

$$r_{21} := \frac{I_2}{I_1} = 0.5110273 \pm 0.0001231.$$

Paper II realizes this ratio by calibrating its effective geometry. Paper III checks conditional compatibility of the quadratic norm. Neither paper currently supplies a value-source theorem selecting r_{21} independently of the gauge profile.

3 Same-scale weak-angle identity

The overlap convention is

$$\frac{1}{g_a^2} = \frac{I_a}{g_{10}^2}.$$

Therefore

$$r_{21} = \frac{I_2}{I_1} = \frac{g_1^2}{g_2^2}. \quad (3.1)$$

Using $g_Y^2 = (3/5)g_1^2$ and the $\overline{\text{MS}}$ definition

$$s_W^2(Q) := \frac{g_Y^2(Q)}{g_Y^2(Q) + g_2^2(Q)},$$

gives the exact identity [1]

$$\boxed{s_W^2(Q) = \frac{3r_{21}(Q)}{5 + 3r_{21}(Q)}}. \quad (3.2)$$

Theorem 3.1 (Gauge-profile round-trip theorem). *If $r_{21}(Q)$ is constructed from the same measured common-scheme pair $(g_Y(Q), g_2(Q))$ through Equation (3.1), then evaluating Equation (3.2) returns the weak angle encoded in that pair. The equality is algebraic and cannot constitute an independent prediction.*

Proof. Substitution of $r_{21} = (5/3)g_Y^2/g_2^2$ into Equation (3.2) gives $g_Y^2/(g_Y^2 + g_2^2)$ identically. $\square$

For the selected ratio,

$$s_W^2(M_t) = 0.2346644 \pm 0.0000433, \quad (3.3)$$

where the uncertainty is propagated from the selected r_{21} row. Direct substitution of the selected g_Y and g_2 gives the same central value. This is a useful convention and arithmetic check, but it has no held-out status.

3.1 Worked example and uncertainty check

As a normalization check, suppose $g_Y = g_2$. Then $r_{21} = 5/3$, and Equation (3.2) gives

$$s_W^2 = \frac{3(5/3)}{5 + 3(5/3)} = \frac{1}{2},$$

exactly as the defining coupling formula does.

For the selected profile, inserting $r_{21} = 0.5110273$ gives $s_W^2 = 0.2346644$. The derivative

$$\frac{ds_W^2}{dr_{21}} = \frac{15}{(5 + 3r_{21})^2}$$

is approximately 0.351443 at that point, so multiplying by the supplied ratio uncertainty 0.0001231 gives 0.0000433. This verifies both the central value and its one-row linear uncertainty propagation.

The interpretation is simple: the numerical agreement confirms the normalization convention and arithmetic transport. It does not create an independent electroweak datum because the same coupling ratio appears on both sides of the calculation.

4 Why the (G_F, m_W) construction remains circular

An electroweak input such as (G_F, m_W) can set an absolute $SU(2)$ normalization, subject to radiative matching. It cannot make the weak-angle test independent when the ratio r_{21} still comes from the measured hypercharge-to- $SU(2)$ profile. Indeed, once g_2 and r_{21} are supplied, Equation (3.1) defines g_1 and hence the weak angle. The information being tested is already present in r_{21} .

At tree level the usual relation is

$$g_2^2 = 4\sqrt{2} G_F m_W^2,$$

with radiative corrections required for a precision common-scheme value. This can supply the overall $SU(2)$ scale, but it supplies no independent value of g_Y/g_2 . The latter is precisely the information already stored in r_{21} .

The earlier tree-level value near 0.2312 used the obsolete 5 TeV profile and a one-loop return to M_Z . It is withdrawn as a prediction. Its threshold scan also showed that the apparent precision was not stable under the stated electroweak matching variation.

5 Criterion for a genuine redundant determination

A non-circular weak-angle test requires all of the following:

1. MTT geometry selects r_{21} without using g_Y , g_1 , g_2 , or $\sin^2 \theta_W$ as value inputs;
2. an absolute gauge normalization and matching scale are selected without using the held-out weak angle;
3. the boundary data are transported with a declared multi-loop scheme and threshold prescription; and
4. the resulting s_W^2 is compared with a measurement not used anywhere in selection, calibration, or branch choice.

Only then is Equation (3.2) a prediction map rather than a round-trip map. Current Papers I–III satisfy the algebraic and realization parts but not the first value-source condition.

6 Conclusion

The weak mixing angle is exactly consistent with the selected gauge profile, as it must be. The precise same-scale result is $s_W^2(M_t) = 0.2346644 \pm 0.0000433$. This validates the hypercharge normalization and overlap-ratio bookkeeping.

It does not add an independent Standard Model observable to MTT closure. The former non-circularity claim fails because the overlap ratio already contains the measured weak-angle information. The next theorem target is sharply defined: select r_{21} from MTT source geometry before consulting the electroweak gauge profile, then execute a held-out common-scheme comparison.

Computational Evidence and Reproducibility

The weak-angle round trip uses the electroweak source row and common-scheme precision workspaces directly, with the global profile audit fixing the comparison convention. Flavor, neutral, HYM, and finite-matrix rows are neighboring context. The non-circular zero-knob source upgrade remains open.

The referenced rows are frozen to the curated results repository state identified below. Hashes are grouped in eight-character blocks for line breaking.

Repository: <https://github.com/PeterNero/mtt-results-repro>
Commit: 31247ebb 5c22f3fb b5443024 365433c6 ee0bff4a
Manifest: release/result_manifest.json
Manifest SHA-256:
fb399689 60b00584 631dbf53 1a708e18
ef928d6b 6d935119 c185d7f6 32b1e7cd

Tier labels are quoted verbatim from that manifest. A row used directly supports only the specific computational statement identified above; a corpus-state cross-check does not prove this paper's local theorems; and an open row is evidence of an unresolved obligation, never of closure.

Rows used directly in this paper.

- A01/current_global_lock (*profile replay*).
Current non-looping global status and source-certificate map.
- A04/final_12_of_12_audit (*profile replay*).
Twelve-obligation embedded renormalized-SM equivalence audit.
- A02/precision_15_source_transport (*profile replay*).
Fifteen measured source coordinates, Jacobian and covariance transport.
- A02/precision_8x8_workspace (*profile replay*).
Eight-coordinate SMDR output with positive-definite 8x8 covariance.
- A01/strict_pew_row (*derived exact*).
Promoted P_EW source row at the declared one-shared-primitive standard.

Corpus-state cross-checks.

- A01/charged_yukawa_higgs_profile (*profile replay*).
Versioned Yu, Yd, Ye and lambda_H profile packet.
- A14/ckm_prediction_profile (*numeric certified*).
Three selected CKM profile rows and uncertainty comparison.

- A01/direct_k_higgs_row (*derived exact*).
Promoted direct K_threshold.Omega_H.lambda row.
- A22/e6_qpsi_qcd_anomaly (*derived exact*).
E6 Qpsi matter/exotic QCD anomaly cancellation audit.
- A19/hym_wiener_contraction (*numeric certified*).
Weighted-theta Fourier-tail and Wiener contraction certificate.
- A40/neutral_two_primitive_profile (*profile replay*).
Measured two-splitting neutrino profile, masses and Dirac Yukawa rows.
- A01/qutrit_weyl_27_matrix (*derived exact*).
Sparse 27x27 qutrit-Weyl left-action realization.

Open boundary (not evidence of closure).

- A05/strict_upgrade_ledger (*open*).
Current 2/9 strict no-knob upgrade ledger.

No imported row changes theorem ownership or promotes a neighboring claim: all local statements retain their stated hypotheses, domains, and limitations.

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